Light

Networks — Basics

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Edexcel GCSE Computer Science — **Networks Basics (1CP2/01 Topic 4: Networks)**. This topic covers why computers are connected in networks, the difference between LANs and WANs, how the internet is structured using IP addresses and routers, the characteristics of wired and wireless connections, network speed calculations, and network topologies.

Exam questions typically ask you to **explain** the purpose of networks, **compare** LAN vs WAN or wired vs wireless, **calculate** transmission times, and **describe** network topologies. Questions range from 1—6 marks, with calculations typically worth 2—4 marks.

Why networks matter

Computer networks are fundamental to modern computing. Almost every digital service we use — from email and web browsing to online gaming and cloud storage — relies on networks. Understanding how networks function helps you

appreciate how data travels from your device to servers around the world and back again.

- **Communication:** Networks enable instant communication through email, messaging, and video calls
- **Resource sharing:** Printers, files, and internet connections can be shared across multiple devices
- **Collaboration:** Teams can work together on shared documents in real-time
- **The modern economy:** Online banking, shopping, and business operations all depend on reliable networks

What you must know

This checklist covers the key exam-visible concepts from the Edexcel specification (4.1.1—4.1.5, 4.1.8).

- **Explain** why computers are connected in networks (resource sharing, communication, centralised management)
- **Define** and **compare** LAN (Local Area Network) and WAN (Wide Area Network)
- **Describe** how the internet is structured using IP addresses and routers
- **Explain** how wired and wireless connectivity characteristics affect performance (speed, range, latency, bandwidth)
- **State** that network speeds are measured in bits per second (kbps, Mbps, Gbps)
- **Calculate** transmission time using: time = file size ÷ transmission rate
- **Describe** characteristics of network topologies: bus, star, and mesh
- **Compare** the advantages and disadvantages of different topologies

> **Specification link:** This topic covers Edexcel 1CP2 specification sections 4.1.1 to 4.1.5 and 4.1.8. Network protocols (4.1.6), the TCP/IP model (4.1.7), and network security (4.2.1) are covered in separate pages.

Toolkit

These are the essential formulas, conversions, and reference tables you need for network calculations.

Network speed units

Unit	Abbreviation	Value in bits per second	Typical use
Bit per second	bps	1 bps	Base unit
Kilobit per second	kbps	1,000 bps	Slow connections
Megabit per second	Mbps	1,000,000 bps	Home broadband
Gigabit per second	Gbps	1,000,000,000 bps	Fast broadband, data centres

Important: Network speeds are measured in **bits** per second, but file sizes are usually measured in **bytes**. Remember: 1 byte = 8 bits. You must convert between them in calculations!

Key formulas

Transmission time = File size (in bits) ÷ Transmission rate (in bps)

Converting file size to bits

If file size is given in bytes:
Size in bits = Size in bytes × 8
Example: 500 KB = 500 × 1,000 × 8 = 4,000,000 bits

Rearranged formulas

File size = Time × Rate
Rate = File size ÷ Time
Always check your units match!

LAN vs WAN comparison

Feature	LAN (Local Area Network)	WAN (Wide Area Network)
Geographic area	Small area (building, campus)	Large area (cities, countries, worldwide)
Ownership	Usually owned by single organisation	Often uses third-party infrastructure
Speed	Higher speeds (typically 1 Gbps+)	Variable, often slower than LAN
Cost	Lower setup and maintenance costs	Higher costs (leased lines, infrastructure)
Example	School network, office network	The internet, company with multiple sites

1. Why use networks?

A **computer network** is two or more computers connected together so they can communicate and share resources. Before networks existed, data had to be transferred using physical media like floppy disks — a slow and inconvenient process sometimes called "sneakernet" (because you had to walk the disk between computers).

1.1 Benefits of networking computers

There are several key reasons why organisations and individuals connect their computers in networks:

Resource sharing

- **Hardware:** Multiple users can share printers, scanners, and storage devices
- **Software:** Applications can be installed centrally and accessed by all users
- **Internet:** A single internet connection can be shared across many devices
- **Files:** Documents can be stored centrally and accessed from any connected device

Communication

- **Email:** Instant message delivery within and outside the organisation
- **Instant messaging:** Real-time text communication between users
- **Video conferencing:** Face-to-face meetings without travel
- **Collaboration tools:** Multiple users can edit documents simultaneously

Centralised management

- **User accounts:** Administrators can manage all user permissions from one location
- **Backups:** Data can be backed up centrally, ensuring nothing is lost
- **Security:** Antivirus and updates can be deployed across all machines
- **Software deployment:** New programs can be installed on all computers at once

Cost savings

- **Shared peripherals:** One high-quality printer serves many users
- **Reduced licensing:** Network software licences can be cheaper than individual ones
- **Efficient storage:** Centralised storage avoids duplication
- **Remote working:** Staff can work from anywhere, reducing office costs

1.2 Drawbacks of networks

While networks offer many benefits, they also introduce some challenges:

- **Security risks:** Networks can be targeted by hackers and malware can spread quickly across connected devices
- **Single point of failure:** If the server or central switch fails, all users may be affected
- **Setup costs:** Initial hardware, cabling, and configuration requires investment
- **Maintenance:** Networks need ongoing management by skilled technicians
- **Dependence:** If the network goes down, users may be unable to work

Worked Example 1 — Benefits of networking

Question: A school is considering connecting all its computers in a network. Give two benefits this would provide. [2 marks]

Answer:
1. Resources such as printers can be shared between all computers, reducing hardware costs. **(1)**
2. Files can be stored centrally on a server, allowing students to access their work from any computer in the school. **(1)**

Exam tip: Always give a brief explanation of why the benefit matters, not just a single word like "sharing."

2. LAN and WAN

Networks are classified by their geographic scope. The two main types you need to know are **LAN** (Local Area Network) and **WAN** (Wide Area Network).

2.1 Local Area Network (LAN)

A **LAN** is a network that covers a small geographic area, typically a single building or campus. The organisation that uses the LAN usually owns and maintains all the hardware and cabling.

Key characteristics of a LAN

- **Small area:** Confined to a single building, floor, or campus
- **Single ownership:** The organisation owns all the network infrastructure
- **High speed:** Typically 1 Gbps or faster using Ethernet
- **Low latency:** Very short delays because of short physical distances
- **Low cost:** No rental fees for communication lines

Examples: School network, home Wi-Fi network, office network

2.2 Wide Area Network (WAN)

A **WAN** covers a large geographic area, connecting LANs that may be in different cities, countries, or continents. WANs often use infrastructure owned by telecommunications companies.

Key characteristics of a WAN

- **Large area:** Spans cities, countries, or the entire world
- **Multiple owners:** Often uses third-party infrastructure (telephone lines, fibre optic cables)
- **Variable speed:** Depends on the connection type and provider
- **Higher latency:** Longer distances mean more delay
- **Higher cost:** Leased lines and infrastructure rental fees

Examples: The internet, a bank with branches across the country, a multinational company's network

Key point: The internet is the largest WAN in the world. It connects millions of smaller networks (including LANs) together using standardised protocols.

Worked Example 2 — LAN vs WAN comparison

Question: Describe two differences between a LAN and a WAN. [4 marks]

Answer:
1. A LAN covers a small geographic area such as a single building **(1)**, whereas a WAN covers a large geographic area such as multiple cities or countries. **(1)**
2. A LAN is typically owned and maintained by a single organisation **(1)**, whereas a WAN often uses infrastructure owned by third parties such as telecommunications companies. **(1)**

Exam tip: For comparison questions, always make a clear contrasting point for each network type.

3. Internet structure

The internet is a global WAN that connects billions of devices worldwide. It relies on two key components: **IP addresses** for identifying devices and **routers** for directing data between networks.

3.1 IP addresses

An **IP address** (Internet Protocol address) is a unique numerical identifier assigned to every device connected to a network. Just as every house has a postal address for mail delivery, every device needs an IP address so data can be sent to the correct destination.

IPv4 addresses

The most common format is **IPv4**, which uses four numbers separated by dots:

```
192.168.1.1
```

- Each number ranges from 0 to 255
- Total of 4 bytes (32 bits)
- Allows approximately 4.3 billion unique addresses
- Running out due to growth of internet-connected devices

IPv6 addresses

IPv6 uses a longer format to provide more addresses:

```
2001:0db8:85a3:0000:0000:8a2e:0370:7334
```

- Uses hexadecimal numbers separated by colons
- Total of 16 bytes (128 bits)
- Provides 340 undecillion unique addresses (enough for every grain of sand on Earth)

- Gradually replacing IPv4 as devices run out of addresses

IP addresses can be **static** (permanently assigned to a device) or **dynamic** (temporarily assigned each time the device connects). Most home devices use dynamic IP addresses, while servers typically use static addresses so they can always be found at the same location.

3.2 Routers

A **router** is a networking device that forwards data packets between different networks. When you send data across the internet, it passes through many routers, each one deciding the best path to the next network until the data reaches its destination.

What routers do

- **Route packets:** Examine the destination IP address and forward data towards its destination
- **Connect networks:** Link your home LAN to your ISP's network and beyond
- **Maintain routing tables:** Store information about available paths through the network
- **Choose best path:** Select the most efficient route for data to travel

How routing works

1. Your device sends a packet with the destination IP address
2. Your router checks its routing table
3. The router forwards the packet to the next router on the path
4. This continues until the packet reaches its destination network
5. The final router delivers the packet to the correct device

Worked Example 3 — Role of routers

Question: Explain the role of a router in a network. [2 marks]

Answer:
A router forwards/directs data packets between different networks. **(1)**
It examines the destination IP address of each packet and determines the best path to send it towards its destination. **(1)**

> **Exam tip:** Mention both "forwarding data" and "between networks" — routers connect networks, not just devices.

Worked Example 4 — Purpose of IP addresses

Question: State the purpose of an IP address. [1 mark]

Answer:
An IP address uniquely identifies a device on a network so that data can be routed to the correct destination. **(1)**

> **Exam tip:** The key word is "unique" — every device needs its own address.

4. Wired vs wireless connectivity

Devices can connect to networks using either physical cables (wired) or radio signals (wireless). Each method has different characteristics that affect network performance.

4.1 Key performance characteristics

When comparing network connections, you need to understand four key characteristics:

Characteristic	Definition	How it affects performance
Speed	The maximum rate at which data can be transmitted	Higher speed = faster file transfers and streaming
Bandwidth	The maximum amount of data that can be transferred in a given time	Higher bandwidth = more users can share the connection
Latency	The delay between sending data and it being received	Lower latency = better for gaming, video calls, real-time applications
Range	The maximum distance over which a connection can operate	Greater range = can cover larger areas without additional equipment

4.2 Wired connections

Wired connections use physical cables to transmit data. The most common type is **Ethernet**, which uses copper cables, though **fibre optic** cables are used for higher speeds and longer distances.

✓ Advantages of wired

- **Faster speeds:** Gigabit (1 Gbps) or higher is common
- **More reliable:** Not affected by interference
- **Lower latency:** Direct physical connection minimises delays
- **More secure:** Harder to intercept without physical access
- **Consistent performance:** Speed doesn't vary with distance (within limits)

✗ Disadvantages of wired

- **Limited mobility:** Devices must be near a network port
- **Installation costs:** Running cables through buildings is expensive
- **Physical constraints:** Cables can be damaged or create trip hazards
- **Less flexible:** Adding new devices requires new cable runs

4.3 Wireless connections

Wireless connections use radio waves to transmit data. **Wi-Fi** is the most common standard for wireless networking in homes and businesses.

✓ Advantages of wireless

- **Mobility:** Users can move around freely within range
- **Easy setup:** No cables to install
- **Flexibility:** Easy to add new devices
- **Cost-effective:** No cabling costs for large areas
- **Convenient:** Can connect from anywhere in range

✗ Disadvantages of wireless

- **Slower speeds:** Generally slower than wired connections
- **Interference:** Walls, other devices, and distance reduce performance
- **Higher latency:** Radio transmission adds delay
- **Less secure:** Radio signals can be intercepted
- **Range limited:** Signal weakens with distance

Worked Example 5 — Wired vs wireless comparison

Question: A company is deciding whether to use wired or wireless networking in their new office. State one advantage and one disadvantage of using wireless instead of wired. [2 marks]

Answer:
Advantage: Employees can move around the office freely with their laptops without being tethered to a network port. **(1)**
Disadvantage: Wireless connections typically have higher latency and may experience interference from walls or other devices, making them less reliable. **(1)**

Exam tip: Relate your answer to the context given. Here, mobility is particularly relevant for an office environment.

5. Network speeds and calculations

Network speeds are measured in **bits per second** (bps). Understanding how to calculate transmission times is essential for exam questions.

5.1 Units of network speed

Network speeds use decimal prefixes (multiples of 1,000), unlike file storage which uses binary prefixes:

Unit	Symbol	Bits per second	Typical connection
Kilobit per second	kbps	1,000 bps	Old dial-up modems (56 kbps)
Megabit per second	Mbps	1,000,000 bps	Home broadband (50—100 Mbps)
Gigabit per second	Gbps	1,000,000,000 bps	Ethernet, fast broadband (1 Gbps)

5.2 The transmission time formula

The key formula for network calculations relates file size, transmission rate, and time:

Time (seconds) = File size (bits) ÷ Transmission rate (bps)

You can rearrange this formula depending on what you need to find:

Find time

Time = Size ÷ Rate

"How long to download?"

Find size

Size = Time × Rate

"How much can I transfer?"

Find rate

Rate = Size ÷ Time

"What speed do I need?"

5.3 Converting units

The most common mistake in network calculations is mixing up bits and bytes. Always convert to consistent units before calculating.

Bytes to bits

File sizes are usually in bytes.

Multiply by 8 to convert to bits.

Example: 5 MB = 5 × 8 = 40 Mb (megabits)

Prefix conversions

1 KB = 1,000 bytes (not 1,024 for network speeds)

1 MB = 1,000 KB = 1,000,000 bytes

1 GB = 1,000 MB = 1,000,000,000 bytes

Worked Example 6 – Calculating transmission time

Question: A file has a size of 240 megabytes (MB). The network has a transmission rate of 80 Mbps. Calculate how long it will take to transfer the file. Show your working. [3 marks]

Step 1: Convert file size from megabytes to megabits:
240 MB × 8 = 1,920 Mb (megabits) **(1)**
Step 2: Apply the formula:
Time = File size ÷ Rate **(1)**
Time = 1,920 ÷ 80 = **24 seconds (1)**

Exam tip: Always show the unit conversion step. This commonly earns a mark even if you make an arithmetic error later.

Worked Example 7 – Working with kilobits

Question: A 5 MB file is being downloaded on a connection with a speed of 500 kbps. Calculate the download time in seconds. [3 marks]

Step 1: Convert MB to bits:
5 MB = 5 × 1,000,000 × 8 = 40,000,000 bits **(1)**
Step 2: Convert kbps to bps:
500 kbps = 500 × 1,000 = 500,000 bps **(1)**
Step 3: Calculate time:
Time = 40,000,000 ÷ 500,000 = **80 seconds (1)**

Exam tip: Convert everything to the base unit (bits and bps) to avoid prefix confusion.

Worked Example 8 – Finding required speed

Question: A video streaming service requires 15 Mb of data to be transferred every second for smooth playback. Calculate the minimum connection speed required in Mbps. [1 mark]

Answer:
Minimum speed = 15 Mb ÷ 1 second = **15 Mbps (1)**

> **Exam tip:** This question is simpler than it looks — the data is already in bits per second format!

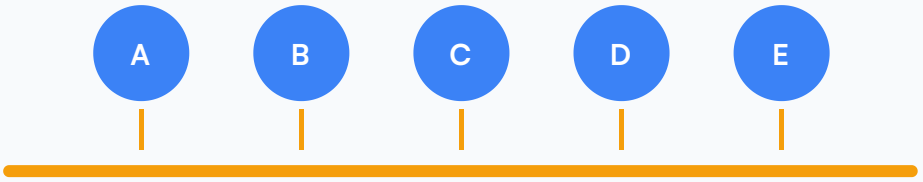
6. Network topologies

A **network topology** describes how devices (nodes) are arranged and connected in a network. The topology affects the network's performance, reliability, and cost. You need to know three topologies: bus, star, and mesh.

6.1 Bus topology

In a **bus topology**, all devices are connected to a single central cable (the "backbone" or "bus"). Data travels along the bus in both directions and each device checks if the data is addressed to it.

Bus Topology



All devices connect to a single backbone cable (shown in orange)

✓ Advantages

- Simple and cheap to install
- Uses less cable than other topologies
- Easy to add new devices
- Works well for small networks

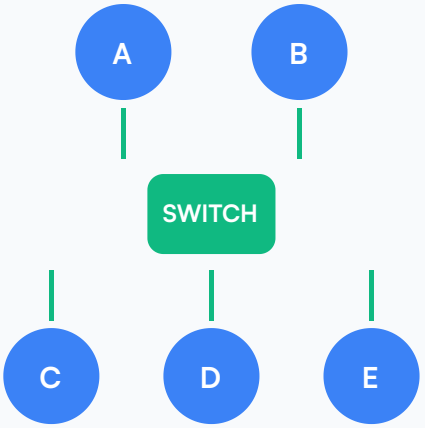
✗ Disadvantages

- If the main cable fails, the whole network fails
- Performance decreases as more devices are added
- Data collisions can occur when multiple devices transmit
- Difficult to troubleshoot faults
- Limited cable length

6.2 Star topology

In a **star topology**, all devices are connected to a central switch or hub. All data passes through this central device, which manages traffic between nodes.

Star Topology



All devices connect to a central switch (connections shown in green)

✓ Advantages

- If one cable fails, only that device is affected
- Easy to add or remove devices
- Easy to detect and isolate faults
- Better performance — no collisions with switches
- Centralised management

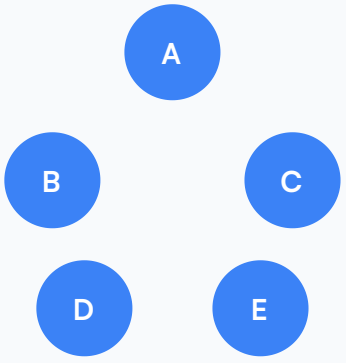
✗ Disadvantages

- If the central switch fails, the whole network fails
- Requires more cable than bus topology
- More expensive due to central switch
- Performance depends on the central device's capacity

6.3 Mesh topology

In a **mesh topology**, each device is connected to multiple other devices. In a **full mesh**, every device connects to every other device. In a **partial mesh**, some devices connect to several others but not all.

Mesh Topology (Full)



In a full mesh, every device has a direct connection to every other device.
This provides redundant paths — if one connection fails, data can use an alternative route.

✓ Advantages

- Very reliable — multiple paths mean no single point of failure
- Data can take alternative routes if one path fails
- Can handle high traffic as load is distributed
- Easy to expand without disrupting the network

✗ Disadvantages

- Expensive — requires many cables and network ports
- Complex to set up and manage
- Full mesh is impractical for large networks
- Troubleshooting can be difficult

6.4 Topology comparison table

Feature	Bus	Star	Mesh
Cable required	Least	Moderate	Most
Cost	Lowest	Moderate	Highest
Single point of failure	Backbone cable	Central switch	None (full mesh)
Fault tolerance	Poor	Moderate	Excellent
Scalability	Limited	Good	Complex
Typical use	Legacy networks	Most LANs	Critical systems, WANs

Worked Example 9 – Describing star topology

Question: Describe the star network topology. [2 marks]

Answer:
In a star topology, all devices are connected to a central switch or hub. **(1)**
All data passes through this central device, which manages traffic between the connected devices. **(1)**

Exam tip: Mention both the physical arrangement (central device) and how data flows (through the centre).

Worked Example 10 – Topology advantages

Question: A hospital is designing a network for its critical patient monitoring systems. State which topology would be most suitable and explain why. [3 marks]

Answer:
A mesh topology would be most suitable. **(1)**
It provides multiple paths for data, so if one connection fails, data can be routed via an alternative path. **(1)**
This high reliability is essential for critical patient monitoring where network failure could be life-threatening. **(1)**

Exam tip: Always relate your choice back to the context — here, reliability is critical for healthcare systems.

Worked Example 11 – Comparing topologies

Question: Explain one advantage of a star topology compared to a bus topology. [2 marks]

Answer:
In a star topology, if one cable fails, only the device connected by that cable is affected. **(1)**
In a bus topology, if the main backbone cable fails, the entire network goes down, affecting all devices. **(1)**

Exam tip: For comparison questions, explain both sides clearly to show why one is better.

Worked Example 12 – Topology disadvantages

Question: A small business has limited budget. Explain why they might choose a bus topology despite its disadvantages. [2 marks]

Answer:
A bus topology uses less cable than star or mesh topologies, reducing installation costs. **(1)**

It also doesn't require expensive central equipment like a switch, making it the cheapest option for a small network with limited budget. **(1)**

Exam tip: Show you understand the trade-offs — cheaper doesn't mean better, but it might be appropriate for the context.

Worked examples summary

The table below summarises the 12 worked examples covered in this topic.

#	Topic	Question type	Marks	Key skill
1	Network benefits	Give benefits	2	Resource sharing, file access
2	LAN vs WAN	Describe differences	4	Geographic scope, ownership
3	Routers	Explain role	2	Forwarding packets between networks
4	IP addresses	State purpose	1	Unique device identification
5	Wired vs wireless	State advantage/disadvantage	2	Mobility vs reliability
6	Transmission time	Calculate with MB and Mbps	3	Convert bytes to bits, apply formula
7	Transmission time	Calculate with kbps	3	Unit conversions, base units
8	Required speed	Calculate minimum rate	1	Rearrange formula
9	Star topology	Describe	2	Central switch, data flow
10	Mesh topology	Justify choice	3	Reliability, context application
11	Star vs bus	Compare advantage	2	Fault tolerance comparison
12	Bus topology	Explain choice	2	Cost considerations

Quick reference

Key terms and definitions

Term	Definition (mark-scheme style)
Network	Two or more computers connected together to share resources and communicate
LAN	Local Area Network; covers a small geographic area such as a building
WAN	Wide Area Network; covers a large geographic area, often using third-party infrastructure
IP address	A unique numerical identifier assigned to each device on a network
Router	A device that forwards data packets between different networks
Bandwidth	The maximum amount of data that can be transmitted over a connection in a given time
Latency	The delay between sending data and it being received at its destination
Topology	The arrangement of devices and connections in a network
Bus topology	All devices connect to a single backbone cable
Star topology	All devices connect to a central switch or hub
Mesh topology	Devices are connected to multiple other devices, providing redundant paths

Network speed quick reference

Conversion	Formula
Bytes to bits	× 8
Bits to bytes	÷ 8
KB to bytes	× 1,000
MB to bytes	× 1,000,000
kbps to bps	× 1,000
Mbps to bps	× 1,000,000
Transmission time	File size (bits) ÷ Rate (bps)

Typical pitfalls

Pitfall 1 — Bits vs bytes confusion.

Wrong: Calculating time using file size in MB and speed in Mbps without converting.
Right: Convert MB to Mb first (multiply by 8), then divide by Mbps.

Pitfall 2 — LAN/WAN definition errors.
Wrong: "A LAN is a small network."
Right: "A LAN covers a small geographic area such as a single building or campus."

Pitfall 3 — Incomplete router description.
Wrong: "A router sends data."
Right: "A router forwards data packets between different networks by examining destination IP addresses."

Pitfall 4 — Topology single point of failure.
Wrong: "Star topology has no single point of failure."
Right: Star topology's single point of failure is the central switch — if it fails, all devices lose connectivity.

Pitfall 5 — Forgetting unit conversions.
Wrong: 100 MB ÷ 50 Mbps = 2 seconds
Right: 100 MB = 800 Mb, then 800 ÷ 50 = 16 seconds

Pitfall 6 — Vague network benefits.
Wrong: "Networks let you share things."
Right: "Networks allow hardware resources such as printers to be shared, reducing costs."

Pitfall 7 — Confusing bandwidth and speed.
Wrong: Using bandwidth and speed interchangeably.
Right: Speed is the rate of data transfer; bandwidth is the maximum capacity. Actual speed may be lower than bandwidth.

Pitfall 8 — Missing context in comparisons.
Wrong: "Wireless is better because of mobility."
Right: "Wireless offers mobility which is beneficial for laptop users in an office, but wired connections would be more reliable for desktop computers."

Pitfall 9 — IPv4 format errors.
Wrong: "An IPv4 address has numbers from 0 to 999."
Right: An IPv4 address has four numbers from 0 to 255, separated by dots (e.g., 192.168.1.1).

Pitfall 10 — Topology comparison incomplete.
Wrong: "Star is better than bus."
Right: Explain why with specific comparison points: "Star is more reliable because one cable failure only affects one device, whereas bus cable failure affects all devices."

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